

10.14 Derive an expression for the magnitude of the force Q required to maintain the equilibrium of the mechanism shown.

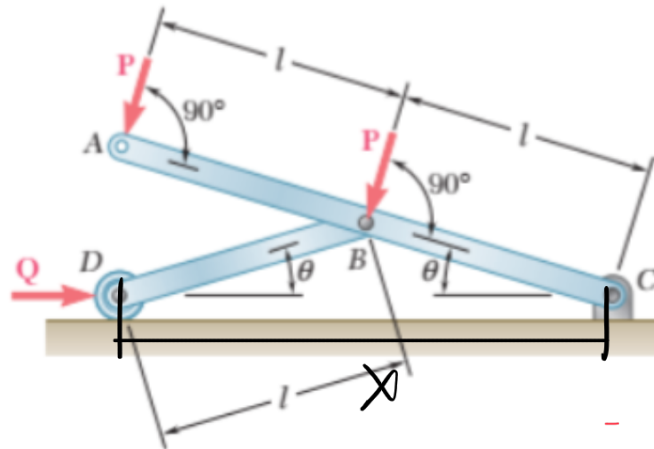


Fig. P10.14

$$x = 2l \cos \theta \quad \checkmark$$

$$A = 2l\theta$$

$$B = l\theta$$

$$\delta x = -2l \sin \theta \delta \theta \quad \checkmark$$

$$\delta A = 2l \delta \theta \quad \checkmark$$

$$\delta B = l \delta \theta \quad \checkmark$$

Virtual work $\delta U = 0 \quad Q \delta x + P \delta A + P \delta B = 0 \quad \checkmark$

$$-2Ql \sin \theta \delta \theta + 2lP \delta \theta + lP \delta \theta = 0$$

$$Q = \frac{3P}{2 \sin \theta} \quad \checkmark$$

10.19 A 4-kN force P is applied as shown to the piston of the engine system. Knowing that $AB = 50$ mm and $BC = 200$ mm, determine the couple M required to maintain the equilibrium of the system when (a) $\theta = 30^\circ$, (b) $\theta = 150^\circ$.

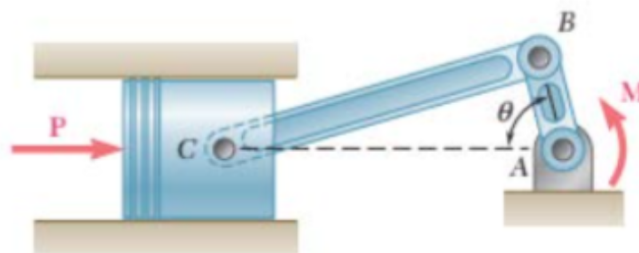
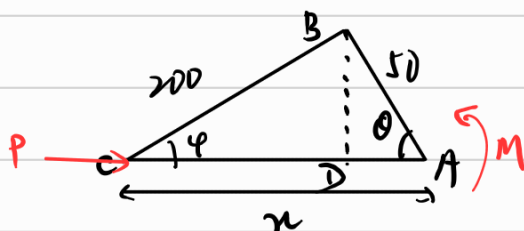


Fig. P10.19 and P10.20

a) $\theta = 30^\circ$



$$BC \cos \varphi + AB \cos \theta = x$$

$$\frac{AB}{\sin \varphi} = \frac{BC}{\sin \theta}$$

$$\delta x = -BC \sin \varphi \delta \varphi - AB \sin \theta \delta \theta$$

$$\sin \varphi = \frac{AB}{BC} \sin \theta$$

$$\delta x = -AB \sin \theta \delta \theta - BC \sin \varphi \left(\frac{AB}{BC} \frac{\cos \theta}{\cos \varphi} \delta \theta \right)$$

$$\cos \varphi \delta \varphi = \frac{AB}{BC} \cos \theta \delta \theta$$

$$\delta x = -\frac{AB}{\cos \varphi} (\sin \theta + \sin \varphi \cos \theta) \delta \theta$$

$$\delta \varphi = \frac{AB}{BC} \frac{\cos \theta}{\cos \varphi} \delta \theta$$

$$\delta x = -\frac{AB \sin(\theta + \varphi)}{\cos \varphi} \delta \theta$$

$$\text{Virtual work } \delta U = -P \delta x - M \delta \theta = 0$$

$$-P \left[-\frac{AB \sin(\theta + \varphi)}{\cos \varphi} \delta \theta \right] - M \delta \theta = 0$$

$$M = \frac{AB \sin(\theta + \varphi)}{\cos \varphi} P$$

When $\theta = 30^\circ$ $P = 4 \text{ kN}$ $BD = 200 \sin \varphi = 50 \sin 30^\circ$
 $\varphi = 7.18^\circ$

$$M = \frac{0.05 \sin(7.18 + 30)}{\cos 7.18} (4000) = 121.82 \text{ N}\cdot\text{m}$$

b) $\theta = 150^\circ$ $M = \frac{0.05 \sin(150 + 7.18)}{\cos 7.18} (4000) = 78.18 \text{ N}\cdot\text{m}$

10.33 A load W of magnitude 600 N is applied to the linkage at B . The constant of the spring is $k = 2.5 \text{ kN/m}$, and the spring is unstretched when AB and BC are horizontal. Neglecting the weight of the linkage and knowing that $l = 300 \text{ mm}$, determine the value of θ corresponding to equilibrium.

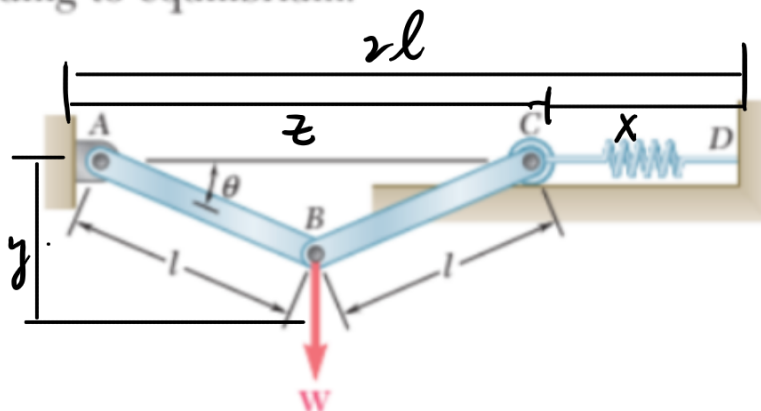


Fig. P10.33 and P10.34

$$z = 2l \cos \theta \quad \checkmark \quad \delta z = -2l \sin \theta \delta \theta \quad \checkmark \quad F = kx = k(2l - z) = 2kl(1 - \cos \theta)$$

$$y = l \sin \theta \quad \checkmark \quad \delta y = l \cos \theta \delta \theta$$

$$\delta U = F \delta z + W \delta y = 0 \quad \checkmark$$

$$2kl(1 - \cos \theta)(-2l \sin \theta \delta \theta) + 600 l \cos \theta \delta \theta = 0 \quad \checkmark$$

$$-4kl^2 \sin \theta \delta \theta (1 - \cos \theta) + 600 l \cos \theta \delta \theta = 0$$

$$-4kl^2 \sin \theta \delta \theta + 4kl^2 \sin \theta \cos \theta \delta \theta + 600 l \cos \theta \delta \theta = 0$$

$$4kl^2 \sin \theta (\cos \theta - 1) = -600 l \cos \theta$$

$$4kl^2 \sin \theta (1 - \cos \theta) = 600 l \cos \theta$$

$$\tan \theta (1 - \cos \theta) = \frac{600}{4(2600)(0.3)} = 0.2 \quad 0.14$$

$$\tan \theta - \sin \theta = 0.2$$

$$\frac{2t}{1-t^2} - \frac{2t}{1+t^2} = 0.2 \quad \frac{4t^3}{1-t^4} = 0.2 \quad 4t^3 = 0.2 - 0.2t^4$$

$$0.2t^4 + 4t^3 = 0.2$$

$$\text{let } t = \tan \frac{\theta}{2}$$

$$\sin \theta = \frac{2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}}{\sin^2 \frac{\theta}{2} + \cos^2 \frac{\theta}{2}}$$

$$= \frac{2 \frac{\sin \theta}{\cos \theta}}{\tan^2 \frac{\theta}{2} + 1}$$

$$= \frac{2t}{1+t^2}$$

$$t_1 = 0.3662$$

$$\frac{\theta}{2} = 20.1127^\circ$$

$$\theta = 40.225^\circ$$

$$t_2 = -20.6001 \quad (\text{reject})$$

10.41 The position of boom ABC is controlled by the hydraulic cylinder BD. For the loading shown, determine the force exerted by the hydraulic cylinder on pin B when $\theta = 65^\circ$.

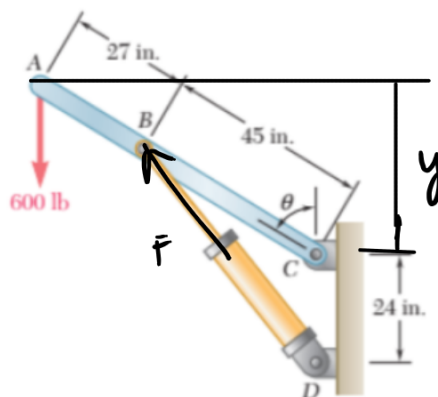


Fig. P10.41 and P10.42

$$y = (45+27)\cos\theta \quad \checkmark \quad \delta y = -(45+27)\sin\theta \delta\theta \quad \checkmark$$

$$BD^2 = BC^2 + CD^2 - 2(BC)(CD)\cos(180^\circ - \theta)$$

$$BD^2 = 45^2 + 27^2 + 2(45)(27)\cos\theta$$

$$BD^2 = 2601 + 2160\cos\theta$$

$$2BD \delta BD = -2160\sin\theta \delta\theta \quad \delta BD = \frac{-1080\sin\theta \delta\theta}{BD} \quad \checkmark$$

$$\delta U = -600\delta y + F\delta BD = 0 \quad \checkmark$$

$$-600 \cdot -(45+27)\sin\theta \delta\theta + F \left(\frac{-1080\sin\theta \delta\theta}{BD} \right) = 0$$

$$43200\sin\theta \delta\theta + \frac{-1080F\sin\theta \delta\theta}{BD} = 0$$

$$BD^2 = 2601 + 2160\cos 65 = 3513.8554$$

$$BD = 59.2778$$

$$F = 43200 \times \frac{59.2778}{1080} = 2371.11 \text{ lb}$$

10.45 The telescoping arm ABC is used to provide an elevated platform for construction workers. The workers and the platform together weigh 500 lb and their combined center of gravity is located directly above C. For the position when $\theta = 20^\circ$, determine the force exerted on pin B by the single hydraulic cylinder BD.

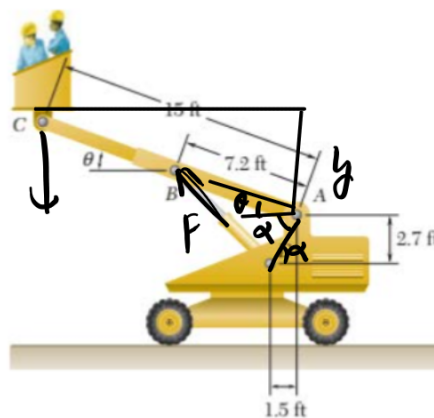


Fig. P10.45

$$AD = 3.0887$$

$$y = 15\sin\theta$$

$$\delta y = 15\cos\theta \delta\theta$$

$$BD^2 = 7.2^2 + AD^2 - 2(7.2)(AD)\cos(\alpha + \theta)$$

$$BD^2 = 61.38 - 44.4773\cos(\alpha + \theta)$$

$$\begin{aligned} & \triangle ADE \\ & \text{AD} = 3.0887 \\ & \text{DE} = 2.7 \\ & \text{AE} = 1.5 \\ & \alpha = \tan^{-1}\left(\frac{2.7}{1.5}\right) \\ & = 60.95^\circ \end{aligned}$$

$$2FD \sin \theta = 44.4773 \sin(\alpha + \theta) \sin \theta$$

$$FD = \frac{22.2387 \sin(\alpha + \theta) \sin \theta}{\sin \theta}$$

$$\delta U = -500 \delta y + F \delta B = 0$$

$$-500(15 \cos \theta \delta \theta) + F \left(\frac{22.2387 \sin(\alpha + \theta) \sin \theta}{FD} \delta \theta \right) = 0$$

The solution procedure
of this problem is
the same as
problem 10.41

$$FD = \sqrt{61.38 - 44.4773 \cos(80.95)} = 7.3745$$

$$F = 7047.67 \text{ lb} \times \frac{7.3745}{22.2387 \sin(80.95)} = 2366.5219 \text{ lb}$$